LESSON 1.6

Math is Ours

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can describe my problem-solving process, name strategies for getting unstuck, and identify the behaviors that make our class a community of math thinkers.

WHY THIS MATTERS What is math? This lesson's answer: math is about DOING — solving problems, getting stuck, trying again, and doing it together. Think about the last time you worked with someone to solve a math problem. What did you actually do?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
representation *Spanish: representación*	A way of showing a problem — a drawing, table, equation, or model — that helps you see it.	The same 12 shown three ways: a tape diagram, a table, and 3 × 4.
strategy *Spanish: estrategia*	A plan of attack for a problem — and something you can switch when you get stuck.	"I will draw a tape diagram first, then divide" — chosen before computing.
critique *Spanish: criticar constructivamente*	To examine an idea and say what is convincing or not — about the idea, never about the person.	"That step assumes the boxes are full — how do we know?"
community agreement *Spanish: acuerdo comunitario*	A rule the whole class agrees to so that everyone can learn together.	"We take turns sharing, and we critique ideas, not people."

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

First I make sense of it: one rack holds 8 bikes, each bike has 2 wheels, so this rack has 8 × 2 = 16 wheels. The other rack has 6 times as many wheels — that is what I don't know yet.

1. I choose a representation: 6 equal groups of 16. Then I make a plan: multiply.
2. 16 × 6 = 96, so the other rack has about 96 wheels. I check my progress: 96 is 6 groups of 16, and 16 × 6 = 96, so my answer matches my plan.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

When we do math, sometimes we get stuck. What can we try?

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ comes before getting stuck because _____, but _____ only helps once _____.
- When I get stuck I can _____ .
- I can also _____ .
- The last time I worked with someone, we _____ .

Word bank representation | strategy | critique | community agreement | make sense of a problem

WATCH OUT!

A common mistake

Treating 'stuck' as a stop sign instead of a signal

LESSON 1.6

Math is Ours

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS What is math? This lesson's answer: math is about DOING — solving problems, getting stuck, trying again, and doing it together. Think about the last time you worked with someone to solve a math problem. What did you actually do?

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

You read a problem and have no idea where to start. According to this lesson, what does that mean?

(A) You are doing math — getting stuck is part of the process, and there are strategies to try

(B) You are not a math person

(C) The problem is broken and should be skipped

(D) You should wait silently until someone gives you the answer

HINT What does the lesson say sometimes happens when we do math?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A classmate shares an idea you think is wrong. What does our community do?

(A) Say 'that's wrong' and move on

(B) Stay silent so nobody's feelings are hurt

(C) Laugh, because wrong answers are funny

(D) Critique the idea respectfully — not the classmate

HINT What do we critique — the person or the idea?

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A bicycle rack holds 8 bikes, and each bike has 2 wheels. How many wheels are on the rack?

(A) 4
(B) 10
(C) 16
(D) 82

HINT How many groups, and how many wheels in each group?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Another bicycle rack has 6 times as many wheels as the rack with 16 wheels. About how many wheels does it have?

(A) About 22
(B) About 66
(C) About 96
(D) About 616

HINT '6 times as many' tells you the operation.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Answer in complete sentences.**

Name TWO strategies from this lesson you can try when you get stuck on a problem.

SENTENCE STARTER When I get stuck I can ___ .
I can also ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

What did you do the last time you worked with someone to solve a math problem?

SENTENCE STARTER The last time I worked with someone, we ___ .
One thing my partner did that helped was ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Compare your two sorts. Which move belongs BEFORE you're stuck and which one only helps AFTER the plan stalls — and how would you tell a classmate which one they need right now?

SENTENCE STARTER ___ comes before getting stuck because ___, but ___ only helps once ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.6

Math is Ours

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

You are solving a problem and realize your strategy is not making progress. What does the problem-solving process say to do?

(A) Keep using the same strategy no matter what

(B) Notice your progress has stalled and shift to a different strategy

(C) Erase everything and decide the problem is impossible

(D) Copy a classmate's paper

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The first rack has 16 wheels and the other rack has 6 times as many. How many BIKES does the bigger rack hold, if every bike has 2 wheels?

(A) 24

(B) 48

(C) 96

(D) 192

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

You are working on your own and get stuck. Which move follows BOTH parts of our working-alone agreements?

(A) Immediately call across the room to a friend

(B) Sit silently for the rest of class without seeking help

(C) Take a classmate's paper to see their work

(D) Try a get-unstuck strategy first, then quietly ask for help without interrupting classmates unnecessarily

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Describe your problem-solving process in order, from reading the problem to checking your answer. Use at least three steps from this lesson.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

What part of problem solving is easy for you? What part is challenging?

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

What skills and knowledge do you bring to a group that help the group be successful?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Compare your two sorts. Which move belongs BEFORE you're stuck and which one only helps AFTER the plan stalls — and how would you tell a classmate which one they need right now?

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LESSON 1.6

Math is Ours

Challenge · Extension

MASTERY CHECK
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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each stuck moment to the strategy that gets you unstuck.

_____ My strategy is not making progress

_____ I don't understand the problem

_____ My answer looks unreasonable

_____ My partner got a different answer

(A) Try a different entry point

(B) Restate it in your own words

(C) Check it against an estimate

(D) Compare steps and find where they differ

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A group solves a hard problem, but one student did all the talking while three stayed silent. By this lesson's standards, how did the GROUP do?

(A) Perfectly — the answer was right, and that is all that matters

(B) Not well — the talker should have stayed silent too

(C) Perfectly — silent members were being respectful

(D) Not well — a community shares thinking and takes turns, so three voices went unheard

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

3 **Answer in complete sentences.**

Write THREE community agreements you would propose for our class — at least one for working together and one for working alone — and for each, say what it protects.

WRITE YOUR ANSWER

4 **Answer in complete sentences.**

Solve this with your full process, narrating each step: a rack has 16 wheels, and another rack has 6 times as many wheels. If a third rack held HALF as many wheels as the big rack, how many bikes would fit on it?

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

A classmate says: 'Math class would be better with no group work — thinking is private.' Using this lesson, write a respectful critique of that IDEA.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Compare your two sorts. Which move belongs BEFORE you're stuck and which one only helps AFTER the plan stalls — and how would you tell a classmate which one they need right now?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help